UPSC EPFO · APFC RECRUITMENT TEST · MODULE 8 · CHAPTER F8.8

Data Interpretation, and Reasoning Essentials

Chapter F8.8 — Three DI Questions Were Asked in 2023, and Exactly One Reasoning Question in 120

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IMPORTANT · THE REASONING DECISION, RESTATED WHERE YOU WILL SEE IT

Blueprint §4.4 records that the 2023 paper contained **three data-interpretation questions** and **exactly ONE pure reasoning question out of 120**.

This chapter is built to that evidence. **Data interpretation gets four sessions and eighteen questions. Reasoning gets one session and seven questions** — despite "General Mental Ability" appearing in the official syllabus, and despite every commercial book for this examination devoting hundreds of pages to syllogisms, blood relations, coding and seating arrangements.

On one question in 120, that material is a **very poor use of your hours**. Register §3.3 records this decision so that it is not silently reversed in a later sitting: **if a future session is tempted to expand reasoning, this is the reason not to** — unless a new paper provides contrary evidence, in which case the Blueprint should be updated first and this chapter second.

What §5 does instead is give you the seven or eight patterns that recover *most* of the value from a single question, in about forty minutes of reading. That is the correct expenditure on one mark of expected value.

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0. How This Chapter Works

STUDY SESSION · CHAPTER F8.8

1. **Session 1** — The DI format and the three-minute rule (§1) ★★★★★
2. **Session 2** — Tables, and the arithmetic of a data set (§2) ★★★★★
3. **Session 3** — Bar charts, line graphs and pie charts (§3) ★★★★☆
4. **Session 4** — The DI trap catalogue (§4) ★★★★★
5. **Session 5** — Reasoning essentials, compact by design (§5) ★★☆☆☆

Then 25 graded questions in §6 — **Q1 to Q18 data interpretation in four sets, Q19 to Q25 reasoning**.

F8.0's calendar takes §6 Q1–12 on Friday of week 3 and Q13–25 on Friday of week 4, works the errors in week 7, and takes all 25 timed in week 11.

NOTE · HOW CHARTS ARE PRESENTED IN A TEXT MANUAL

This manual is generated from Markdown, so a bar chart or pie chart cannot be drawn on the page. The practice sets in §6 therefore give you the **values as they would be read off the chart**, in a table.

That is not the compromise it looks like, because **converting a chart into a small table is exactly what you should do in the exam hall anyway** — see §3.2. The reading of the chart is a fifteen-second mechanical step; the marks are in the arithmetic that follows, and that is what §6 drills.

Where a question depends on a feature only a picture carries — the visual steepness of a line, the relative size of two sectors — §3 tells you what to look for and §6 supplies the numbers that feature would give you.

1. Session 1 — The DI Format ★★★★★

1.1 What it is, and why it is the best trade in the Quant block

A single data set — a table, a bar chart, a pie chart or a line graph — followed by **three to five questions**. In 2023 it was **three questions on one set** (PYQ-DERIVED, Blueprint §4.4).

MUST MASTER · THREE MINUTES FOR A SET OF THREE, AND WHY THE SET IS CHEAP

F8.0 §1.1 budgets **3 minutes for a set of three DI questions** — one minute each, which is the paper's average, yet DI feels slower than average. The reason it still fits is that **the cost of reading the data is paid once and serves every question in the set**.

This makes DI structurally the same trade as reading comprehension (F8.3 §0): a fixed cost amortised over several marks. And it produces the same rule:

> **Spend twenty seconds understanding the data before you read the first question. Then answer all of them without going back to first principles.**

DI is the only place in the Quant paper where no new knowledge is needed. There is no formula to misremember, no theorem, no trap of a matched pair. Everything you need is printed on the page. The questions are percentage, ratio and average — F8.5's material — applied to numbers you are given.

For a candidate who is weak in mensuration or algebra, **this is the block to secure first.**

1.2 The twenty seconds, spent properly

Before reading any question, establish four things:

Check	Why it matters
What exactly does each row and column measure?	Confusing *claims received* with *claims settled* loses every question in the set
What are the UNITS?	Thousands? Lakhs? Crores? Percentages of something?
Is any figure a total or a subtotal?	Totals are often given; computing one you were given wastes 30 seconds
Is any column a percentage rather than a quantity?	This is the source of the trap in §4.1

Then compute the row and column totals you are likely to need — usually the grand total, because most questions ask for a share of it. In §6.1 the three annual totals (1,000, 1,100 and 1,320) answer or part-answer four of the five questions in the set.

CHECK YOURSELF · SESSION 1

1. What is the time budget for a set of three DI questions, and why does it fit?
2. What four things do you check before reading the first question?
3. Why is DI the block to secure first if your algebra is weak?

2. Session 2 — Tables ★★★★★

The table is the purest form of the format, and every other form reduces to it.

2.1 The five calculations that cover almost every DI question

Asked	Method
Total, or a share of the total	Sum the row or column; share = part ÷ total × 100
Percentage increase or decrease	(change ÷ **the earlier value**) × 100
Ratio of two entries	Divide, then reduce
Average over a period	Sum ÷ the number of periods
Which is highest or lowest	Compare **ratios**, not differences, when the question says *rate* or *per cent*

2.2 Comparing growth without computing every percentage

MUST MASTER · YOU RARELY NEED TO CALCULATE ALL FOUR PERCENTAGES

When a question asks which of several items grew fastest, computing four percentages takes ninety seconds you do not have. Two shortcuts:

1. Compare the fraction, not the percentage. Growth from 240 to 300 is 60/240 = **1/4**; from 320 to 352 is 32/320 = **1/10**. You can see 1/4 beats 1/10 without converting either to a percentage.

2. Eliminate by inspection first. An item that grew by a small absolute amount **from a large base** cannot be the fastest grower. In §6.1's table, R rose 32 on a base of 320 and S actually **fell** — both are out in five seconds, leaving only P and Q to compute.

The examiner's counter-trap: the item with the **largest absolute increase** is often not the item with the largest **percentage** increase, and both are offered. In §6.1, P rose by 60 and R by 32, but the question is about rate, so read the stem for the word *percentage* or *rate* and let it choose your arithmetic.

2.3 The approximation habit

DI options are usually far apart, so **round aggressively and check the gap**. 443/540 does not need long division: 540 × 0.8 = 432 and 540 × 0.85 = 459, so the answer lies between 80% and 85%, and if the options are 78, 82, 85 and 88, **82% is the only candidate** — Q9.

Round to the nearest convenient figure, get the range, then look at the options. If two options fall inside your range, only then compute precisely.

CHECK YOURSELF · SESSION 2

1. Name the five calculations that cover most DI questions
2. On which value is a percentage change always based?
3. Give the two shortcuts for finding the fastest grower
4. Estimate 443/540 without dividing

3. Session 3 — Charts ★★★★☆

3.1 What each chart form is for

Form	Shows	Watch for
Bar chart	Comparison of quantities across categories	Whether bars are **grouped** (two series) or **stacked** (parts of a whole)
Line graph	A trend over time	**Steepness is the rate of change**; a falling line can still be above another rising one
Pie chart	**Shares of one whole** — never absolute quantities unless the total is stated	The total must be given for any rupee answer
Stacked bar	Both the total and its composition	A segment can grow while its **share** falls

3.2 Pie charts — degrees, percentages and amounts

MUST MASTER · 360° IS THE WHOLE, SO 3.6° IS ONE PER CENT

Every pie chart question is one of these three conversions:

> **percentage = (central angle ÷ 360) × 100 · central angle = percentage × 3.6 · amount = percentage × total**

Angle	Share	Angle	Share
36°	10%	120°	33⅓%
54°	**15%**	**144°**	**40%**
72°	20%	180°	50%
90°	**25%**	240°	66⅔%

And the rule that matters most: a pie chart alone can never give you a rupee figure. It gives shares. If the question wants an amount, the total must be stated somewhere in the stem — and if two pie charts are shown with **different totals**, their sectors cannot be compared directly at all. A 25% sector of ₹8 crore is larger than a 40% sector of ₹4 crore.

That last point is the single most productive trap in pie-chart questions, and §4.3 is its general form.

3.3 Reading a chart into a table — the fifteen-second step

In the exam hall, **write the numbers down**. Copy the bar heights or sector percentages into a small table in the margin before answering anything. Reasons:

1. You will otherwise **re-read the chart** for every question, at ten seconds a time
2. Reading a value off a picture is where the error occurs; doing it once, deliberately, halves the risk
3. The totals you need are far easier to compute from a written column than from a picture

This is why §6's sets are given as tables: **the table is the state you should convert every chart into.**

CHECK YOURSELF · SESSION 3

1. What percentage corresponds to 54°? 144°? What angle to 25%?
2. Why can a pie chart alone never give an amount?
3. When can two pie charts' sectors not be compared?
4. What is the first thing to do on meeting a bar chart?

4. Session 4 — The DI Trap Catalogue ★★★★★

DI questions are not conceptually hard, so the examiner's marks come almost entirely from these five confusions.

4.1 Percentage against percentage point

MUST MASTER · A RATE THAT MOVES FROM 12% TO 9% HAS FALLEN THREE PERCENTAGE POINTS, NOT THREE PER CENT

This is the most examined distinction in data interpretation anywhere, and it is worth learning as a sentence you can recite.

- The **percentage-point** change is the simple difference: 12 - 9 = **3 percentage points**
- The **percentage** change is relative to the starting rate: 3/12 = **25%**

Both are correct answers to **different** questions, and both will be on the paper — Q16. So read the stem precisely:

Stem says	Answer
"fell by how many percentage points"	**3**
"the rate fell by what per cent"	**25%**
"the default rate fell from ... to ..."	**3 percentage points** — the units of a rate difference

The general habit: when a quantity is itself a percentage, ask whether the question wants the difference of the percentages or the percentage of the difference.

4.2 Absolute against relative

The largest **number** and the largest **share** are different questions, and they frequently have different answers. In §6.4's table, 2023 has the most establishments covered but the **lowest** default rate, while 2020 has the most **defaulters** despite having far fewer establishments than 2023 — Q17.

Where a question asks "highest" or "most", find the noun. *Most defaulting establishments* is a count; *highest rate of default* is a ratio.

4.3 The wrong base

The single commonest arithmetic error in DI. A rise from 500 to 800 is **60%** — the base is the earlier figure, 500 — and **not** 37.5%, which is 300/800 and answers no question at all. Q18 offers both.

Say the base out loud before dividing: *"per cent of what?"* This is F8.5 §1.4's rule, and it accounts for more lost DI marks than every other error combined.

4.4 The changed total

A share can fall while the amount rises, if the total is growing. If establishment expenditure is 40% of ₹7.2 crore and remains 40% of ₹9 crore, the **share is unchanged** but the amount has risen from ₹2.88 crore to ₹3.6 crore — Q14.

So: **when a question changes the total, recompute the amount; when it changes a share, do not assume the total moved.**

4.5 The unit

Thousands, lakhs and crores are mixed deliberately, and a table headed *"in thousands"* will have a question whose options are in lakhs. **Write the unit next to your answer** before comparing with the options. Q7's answer is 24 **thousand**, and Q13's is ₹36 **lakh** from a table given in crores.

> **THINK LIKE UPSC · WHY DI REWARDS SLOWNESS AT ONE POINT AND SPEED EVERYWHERE ELSE**
>
> Notice what all five traps have in common: **none of them is a computational failure.** Every one is a failure to read precisely — the wrong base, the wrong noun, the wrong unit, the wrong kind of change.
>
> That produces an unusual instruction. In DI you should be **fast at the arithmetic and slow at the stem** — which is the reverse of what candidates naturally do, because the arithmetic feels like the hard part and so absorbs the attention.
>
> **Read the stem twice. Compute once, roughly.** That is the whole discipline of the format, and it is also Blueprint §8's remedy for the *"which one is not correct"* pattern applied to numbers.

5. Session 5 — Reasoning Essentials ★★★★★

One question in 120. What follows is therefore the complete treatment: the patterns that recur, at one worked line each. Read it once, do Q19 to Q25, and do not return to it until week 11.

5.1 Number and letter series

Compute the **differences**. If they are not constant, compute the differences **of the differences**.

> 3, 6, 11, 18, 27, ? — differences 3, 5, 7, 9, so the next difference is 11 and the answer is **38** — Q19.

Other patterns worth a glance: alternating series (two interleaved sequences), multiply-and-add (×2 then +1), and squares or cubes displaced by one (2, 5, 10, 17 is $n^2 + 1$).

5.2 Coding and decoding

Compare the coded word letter by letter with the original and find the **shift**.

> EPFO » FQGP is **+1** on every letter, so RTI » **SUJ** — Q20.

Shifts may be negative, may alternate (+1, -1, +1), or may reverse the word. **Write the alphabet with its position numbers** in the margin if more than one shift is involved; it takes ten seconds and prevents the only real error in the topic.

5.3 Blood relations

> **KNOW · RESOLVE THE PHRASE FROM THE INSIDE OUT**
>
> Take the longest phrase apart from its innermost term outwards, substituting a plain word at each step.
>
> > *"the daughter of my grandfather's only son"* » **grandfather's only son = my father** » *"the daughter of my father"* = **my sister** — Q21.
>
> Two cautions the examiner relies on: **"only son"** is the key that collapses a generation, and the answer is often a **sister or brother** where candidates expect a cousin. Also note that questions rarely specify gender for in-laws, so a relation like *"brother-in-law"* may be unresolvable — in which case the option saying *cannot be determined* is correct.

5.4 Direction sense

Draw it. Take north as up, and remember that **a right turn while facing north is east, while facing south is west** — reversing after a turn is where errors happen.

> 5 km north, right 3 km, right 5 km: the two 5 km legs cancel, leaving **3 km east** — Q22.

Net displacement is found by cancelling opposite legs and, where both a north–south and an east–west remainder survive, by Pythagoras (F8.6 §5.1's triples appear here too).

5.5 Syllogisms

Only two rules are needed at this level:

1. **From two premises, a conclusion follows only if it MUST be true** — the same test as F8.3 §3.2's inference rule
2. **"Some A are B" always yields "some B are A"** — a universal *all A are B* likewise yields *some B are A*. But **"all A are B" and "some B are C" yields nothing** about A and C

> *All auditors are graduates. Some graduates are lawyers.* Does *some auditors are lawyers* follow? **No** — the lawyers among the graduates need not be the auditors. Does *some graduates are auditors* follow? **Yes**, by rule 2 — Q23.

The commonest error is accepting a conclusion because it is plausible in the world. Premises are to be taken as given, however false, and nothing outside them may be used.

5.6 Odd one out, and ranking

Odd one out: test the obvious classes in order — perfect squares, cubes, primes, multiples of a number, even against odd. *121, 144, 169, 150*: the first three are 11^2, 12^2 and 13^2, so **150** is odd — Q24.

Ranking and ordering: write a single line with the seniormost at the left and insert each person as the clues arrive.

> *C is senior to A but junior to B; D is the most junior; E is senior to A but junior to C.* Building it: B > C, then C > E > A, then D last, giving **B > C > E > A > D**, so the most senior is **B** — Q25.

CHECK YOURSELF · SESSION 5

1. Continue: 2, 5, 10, 17, 26, ?
2. If DELHI is coded CDKGH, how is PATNA coded?
3. *"My mother's brother's only sister"* is who?
4. Does *some A are B* imply *some B are A*? Does *all A are B, some B are C* imply anything about A and C?
5. Walk 4 km east, turn left, 3 km, turn left, 4 km. Where are you?

6. Graded Practice — 25 Questions ★★★★★

Q1 to Q18 are data interpretation, in four sets. Q19 to Q25 are reasoning. Target **3 minutes per DI set of three to five**, and **40 seconds** per reasoning question.

6.1 Set I — Table: contributions received by four regional offices (₹ crore)

Office	2021	2022	2023
P	240	300	330
Q	180	198	270
R	320	352	396
S	260	250	324

Q1. The total contribution received by all four offices in 2022 was (a) ₹1,000 crore (b) ₹1,100 crore (c) ₹1,150 crore (d) ₹1,320 crore

Q2. The office recording the highest **percentage** increase in contributions from 2021 to 2022 was (a) Q (b) R (c) S (d) P

Q3. The contribution of office R in 2023, as a percentage of the total contribution of that year, was nearest to (a) 30% (b) 33% (c) 27% (d) 36%

Q4. The average annual contribution of office Q over the three years was (a) ₹198 crore (b) ₹210 crore (c) ₹216 crore (d) ₹222 crore

Q5. The percentage increase in the total contribution of all four offices from 2022 to 2023 was (a) 17% (b) 20% (c) 22% (d) 32%

6.2 Set II — Bar chart, read into a table: claims received and settled (thousands)

Year	Claims received	Claims settled
2019	80	64
2020	90	63
2021	100	85
2022	120	96
2023	150	135

Q6. The year in which claims settled formed the **lowest** percentage of claims received was (a) 2020 (b) 2019 (c) 2022 (d) 2021

Q7. The number of claims received but not settled in 2022 was (a) 16 thousand (b) 20 thousand (c) 22 thousand (d) 24 thousand

Q8. The percentage of claims settled in 2023 was (a) 80% (b) 85% (c) 90% (d) 95%

Q9. Taking the five years together, the percentage of claims received that were settled was nearest to (a) 82% (b) 78% (c) 85% (d) 88%

Q10. The percentage increase in claims received from 2019 to 2023 was (a) 70% (b) 87.5% (c) 47% (d) 100%

6.3 Set III — Pie chart: an office's annual expenditure of ₹7.2 crore

Head of expenditure	Central angle
Establishment	144°
Pension disbursement	90°
IT and systems	54°
Rent	36°
Other charges	36°

Q11. The amount spent on establishment was (a) ₹1.80 crore (b) ₹2.16 crore (c) ₹2.60 crore (d) ₹2.88 crore

Q12. The sector representing pension disbursement corresponds to what percentage of total expenditure? (a) 15% (b) 20% (c) 25% (d) 30%

Q13. Expenditure on IT and systems exceeded expenditure on rent by (a) ₹36 lakh (b) ₹18 lakh (c) ₹54 lakh (d) ₹72 lakh

Q14. If total expenditure in the following year rose by 25% while the **share** of establishment remained unchanged, the amount spent on establishment would be (a) ₹3.24 crore (b) ₹3.60 crore (c) ₹2.88 crore (d) ₹4.50 crore

6.4 Set IV — Establishments covered, and those in default (thousands)

Year	Establishments covered	Of which in default
2019	500	60
2020	550	77
2021	600	66
2022	700	70
2023	800	72

Q15. The rate of default in 2020 was (a) 11% (b) 12% (c) 13% (d) 14%

Q16. Between 2019 and 2023 the rate of default fell by (a) 3% (b) 25% (c) 3 percentage points (d) 12 percentage points

Q17. The year in which the **number** of establishments in default was highest was (a) 2020 (b) 2023 (c) 2022 (d) 2021

Q18. The number of establishments covered in 2023 exceeded that in 2019 by (a) 300% (b) 60% (c) 37.5% (d) 62.5%

6.5 Reasoning essentials

Q19. Complete the series: 3, 6, 11, 18, 27, ? (a) 34 (b) 36 (c) 38 (d) 40

Q20. If EPFO is coded as FQGP, then RTI is coded as (a) QSH (b) SVK (c) TUJ (d) SUJ

Q21. Pointing to a photograph, a man said, "She is the daughter of my grandfather's only son." How is she related to him? (a) His sister (b) His daughter (c) His cousin (d) His niece

Q22. A man walks 5 km towards the north, turns right and walks 3 km, then turns right and walks 5 km. He is now (a) 3 km north of the starting point (b) 3 km east of the starting point (c) 5 km east of the starting point (d) 8 km east of the starting point

Q23. Premises: All auditors are graduates. Some graduates are lawyers. **Conclusions:** I. Some auditors are lawyers. II. Some graduates are auditors. Which conclusion follows? (a) Only I (b) Both I and II (c) Only II (d) Neither I nor II

Q24. Choose the odd one out. (a) 121 (b) 144 (c) 169 (d) 150

Q25. Five officers A, B, C, D and E are ranked by seniority. C is senior to A but junior to B. D is the most junior of all. E is senior to A but junior to C. The most senior officer is (a) B (b) C (c) A (d) E

7. Answers and Explanations

7.1 Set I

Q	Ans	Explanation
1	b	300 + 198 + 352 + 250 = **₹1,100 crore**. Compute all three annual totals at the outset — 1,000, 1,100 and 1,320 — because they serve Q3 and Q5 as well. §1.2
2	d	**P**: 60 on 240 = 25%. Q: 18 on 180 = 10%; R: 32 on 320 = 10%; S actually **fell**. Note that R had the second-largest absolute rise but only 10% growth — the absolute-against-relative trap of §4.2
3	a	396/1,320 = **30%**. Estimate rather than divide: 10% of 1,320 is 132, and 3 × 132 = 396 exactly
4	c	(180 + 198 + 270)/3 = 648/3 = **₹216 crore**. (a) is the middle year's figure, which is not the mean
5	b	(1,320 - 1,100)/1,100 = 220/1,100 = **20%**. (d) 32% is the increase from **2021**, the wrong base year — §4.3, and the reason to read which two years the stem names

7.2 Set II

Q	Ans	Explanation
6	a	Rates are 80%, **70%**, 85%, 80%, 90%, so **2020**. The question asks for a **percentage**, so compare ratios; 2020 also has fewer settled claims in absolute terms than later years, but that is not what decides it
7	d	120 - 96 = **24 thousand**. Note the unit: the table is in thousands, so the answer is 24,000 claims — §4.5
8	c	135/150 = **90%**
9	a	Received 540, settled 443. Estimate: 80% of 540 = 432 and 85% = 459, so the answer lies between, and only **82%** is offered in that range — §2.3. No long division needed
10	b	(150 - 80)/80 = 70/80 = **87.5%**. (a) 70 is the **absolute** increase in thousands, offered as a percentage — the single most common DI slip

7.3 Set III

Q	Ans	Explanation
11	d	144/360 = 40%, and 40% of ₹7.2 crore = **₹2.88 crore**. (a) is 25% of the total, the pension figure
12	c	90° is a quarter of 360°, so **25%**. Reading a right angle as a quarter is instant and needs no arithmetic
13	a	54° - 36° = 18°, which is 5% of the total: 5% of ₹7.2 crore = ₹0.36 crore = **₹36 lakh**. The table is in crores and the options in lakhs — §4.5. (b) 18 lakh mistakes the 18° for a percentage
14	b	The new total is 7.2 × 1.25 = ₹9 crore, and 40% of 9 = **₹3.60 crore**. The **share** is unchanged, the **amount** is not — §4.4. (c) is the old amount, for the candidate who read "unchanged" as applying to the money

7.4 Set IV

Q	Ans	Explanation
15	d	77/550 = **14%**. Check by parts: 10% of 550 is 55, and 4% is 22, totalling 77
16	c	The rate fell from 12% to 9%, a fall of **3 percentage points**. (b) 25% is the *relative* fall, 3/12 — a correct answer to *"by what per cent did the rate fall"*, but the stem asks by how much the rate fell, and a difference of rates is measured in percentage points. §4.1 — the most examined distinction in DI
17	a	**2020, with 77 thousand** — more than 2023's 72 thousand, even though 2023 covers half again as many establishments. The stem says **number**, not rate; the rate in 2020 was the worst too, but the point of the question is that the counts and the rates rank differently. §4.2
18	b	(800 - 500)/500 = **60%**. (c) 37.5% is 300/800, the wrong base; (a) 300% mistakes the absolute difference for a percentage. §4.3

7.5 Reasoning

Q	Ans	Explanation
19	c	Differences 3, 5, 7, 9 — the odd numbers — so the next is 11 and 27 + 11 = **38**. (b) 36 assumes a constant difference of 9
20	d	Each letter advances by one: E»F, P»Q, F»G, O»P. So R»S, T»U, I»J = **SUJ**. (c) leaves the first letter unchanged, (a) shifts backwards
21	a	*Grandfather's only son* = **his father**; the daughter of his father is **his sister**. (c) is the answer if you overlook the word *only*, which is precisely why it is there
22	b	North 5, then east 3, then south 5. The two vertical legs cancel exactly, leaving **3 km east**. Draw it — the second right turn from an easterly heading is southward, not northward
23	c	**Only II.** *Some graduates are auditors* is the valid converse of *all auditors are graduates*. Conclusion I does **not** follow: the lawyers among the graduates need not include any auditor. Plausibility is not entailment — §5.5, and the same test as F8.3's inference rule
24	d	$121 = 11^2$, $144 = 12^2$, $169 = 13^2$. **150** is not a perfect square
25	a	B > C from "C is junior to B"; C > E > A from the last clue with "C is senior to A"; D is last. The order is **B > C > E > A > D**, so the most senior is **B**

THINK LIKE UPSC · WHAT THIS BANK WAS ACTUALLY TESTING

1. **The base.** Q5, Q10, Q18. **Three of eighteen DI questions**, and in each the wrong option is a real number computed on the wrong denominator. *Per cent of what?* — asked out loud — is worth more marks in DI than any technique
2. **Percentage against percentage point.** Q16, where **both** answers are on the paper and only the stem distinguishes them
3. **Count against rate.** Q2, Q6, Q17. The item with the biggest number is repeatedly not the item with the highest rate, and the question alternates between asking for each
4. **The unit.** Q7 and Q13 — thousands, and a table in crores with options in lakhs
5. **Estimation as a method, not a shortcut.** Q3 and Q9 are both solved faster by bracketing than by dividing. In a 120-minute paper, arithmetic you avoid is arithmetic you cannot get wrong

And note the shape of the reasoning half: **seven questions covering seven distinct patterns**, one each. That is the right coverage for **one question in 120** — enough to recognise whatever appears, not enough to have cost you a week.

8. One-Minute Revision

ONE-MINUTE REVISION · CHAPTER F8.8

WEIGHT 3 DI questions in 2023 · exactly ONE pure reasoning question in 120 · DI budget **3 minutes per set of three** · reasoning 40 seconds each

WHY DI IS CHEAP the cost of reading the data is paid **once** and serves the whole set · **no new knowledge is needed** — it is percentage, ratio and average applied to printed numbers. Secure this block first if algebra is weak

THE TWENTY SECONDS FIRST what does each row measure · **what are the units** · what is a total · which column is a percentage · **then compute the grand total**

THE FIVE CALCULATIONS total and share · **percentage change on the EARLIER value** · ratio · average · highest by **ratio** when the stem says *rate* or *per cent*

SHORTCUTS compare **fractions** not percentages (60/240 = 1/4 beats 32/320 = 1/10) · eliminate a small rise on a large base by inspection · **round, bracket, then look at the options** — 443/540 is between 80% and 85%

PIE CHARTS 3.6° = 1% · 36° = 10% · **54° = 15%** · 72° = 20% · **90° = 25%** · 120° = 33⅓% · **144° = 40%** · 180° = 50% · **a pie chart alone gives shares, NEVER amounts** · two pies with different totals **cannot** be compared sector to sector

CHARTS grouped bars = two series, stacked = parts of a whole · **line steepness = rate of change** · a segment can grow while its **share** falls · **copy every chart into a table before answering**

THE FIVE TRAPS 1 percentage vs **percentage point** — 12% to 9% is **3 percentage points**, or a **25%** fall · **2** absolute vs relative — biggest number is not the highest rate · **3 the wrong base** — 500 to 800 is 60%, not 37.5% · **4** the changed total — share unchanged, amount risen · **5** the unit — thousands, lakhs, crores

BE SLOW AT THE STEM AND FAST AT THE ARITHMETIC. Every DI trap is a reading failure, not a computation failure

REASONING, COMPACTLY series » take **differences, then differences of differences** · coding » find the **shift**, write the alphabet out · blood relations » resolve **from the inside out**, watch **"only son"** · directions » **draw it**, cancel opposite legs · syllogism » must follow, not plausible; **some A are B gives some B are A**, but *all A are B + some B are C* gives **nothing** · odd one out » test squares, cubes, primes, multiples · ranking » one line, seniormost on the left

9. Five-Minute Wall Sheet

FIVE-MINUTE WALL SHEET · CHAPTER F8.8

READ THE DATA ONCE, PROPERLY. IT PAYS FOR THREE TO FIVE QUESTIONS

WRITE THE UNITS DOWN. THOUSANDS? LAKHS? CRORES?

COMPUTE THE GRAND TOTAL BEFORE THE FIRST QUESTION

PERCENTAGE CHANGE IS ON THE *EARLIER* VALUE. 500 » 800 IS 60%, NOT 37.5%

"PER CENT OF WHAT?" — SAY IT OUT LOUD BEFORE DIVIDING

12% TO 9% = 3 PERCENTAGE POINTS = A 25% FALL. THE STEM CHOOSES

BIGGEST NUMBER IS NOT HIGHEST RATE. FIND THE NOUN

3.6° = 1% · 54° = 15% · 90° = 25% · 144° = 40%

A PIE GIVES SHARES, NOT RUPEES — UNLESS THE TOTAL IS STATED

SHARE UNCHANGED + TOTAL UP 25% = AMOUNT UP 25%

ROUND AND BRACKET. IF ONE OPTION IS IN RANGE, THAT IS THE ANSWER

COMPARE FRACTIONS, NOT PERCENTAGES

SLOW ON THE STEM, FAST ON THE SUM

SERIES » DIFFERENCES, THEN DIFFERENCES OF DIFFERENCES

CODING » FIND THE SHIFT · RELATIONS » INSIDE OUT · DIRECTIONS » DRAW IT

SYLLOGISM: MUST FOLLOW, NOT PLAUSIBLE. *SOME A ARE B* » *SOME B ARE A*

ONE LINE FOR THE INTERVIEW Data interpretation is the part of an aptitude paper that most resembles the work itself — the numbers are supplied and the skill being tested is reading them precisely, which is why almost every error in the format is a misread base or a misread unit rather than a failure of arithmetic.

ACTION · MODULE 8 IS COMPLETE

Nine chapters, **F8.0 to F8.8**, covering the largest single block in the paper — **80 planning marks** — and every format observed in the 2023 paper: 10 vocabulary, 5 para jumbles, 5 comprehension, 15 Quant across arithmetic, mensuration, algebra, probability and DI, and the single reasoning question.

But this module is not finished by being read. It is finished by being **run**, thirty minutes an evening for twelve weeks, on F8.0's calendar. Go to **F8.0 §3, week 1, Monday**, and start tomorrow.

Next in the manual: Module 4 — Indian Economy, Population, Development and Globalisation. Five chapters, 8 hours, 15 marks — the leanest content module left, and per Blueprint §11.3 the first place to sacrifice depth if work eats a week.

Before you leave Module 8, state without looking: the fraction for $16\frac{2}{3}\%$; the curved and total surface areas of a hemisphere; what $\alpha + \beta$ and $\alpha\beta$ equal; how many face cards are in a pack; and the difference between a percentage and a percentage point.